

Influencer-Brand Matching via GAIL & Graph Spillover — Capstone Presentation

1. Project Overview / Abstract / Scope (2 slides)

1.1 Project Overview

- **Problem:** Brands choose influencers on follower count / gut feel; no honest measure of *collaboration spillover* (how an influencer's association with other creators/brands lifts engagement-per-rupee).
- **Goal:** End-to-end pipeline — ingestion → graph → GAIL spillover → fusion (spillover + sentiment risk + creator features) → brand query → ranked recommendations with honest 0–100 score + confidence + explainability, surfaced in a web app.
- **Thesis:** *Better data, better model* — honest small-N uncertainty (N=10 effective labeled, CI ± 13 –21 pts) beats over-confident point estimates (CAPSTONE_NEXT_STEPS.md : 60,795).

1.2 Scope & Constraints

- **Stack:** Supabase (Postgres + pooler `aws-0-ap-south-1`), Python (FastAPI `backend/` , PyG HeteroData `ml/`), Next 16 + Tailwind v4 `frontend/` .
 - **Tracks:** A Data/Infra → facts, C Fusion+Backend → edges/fusion/API, B ML-Core → graph/GAIL, D Frontend+App → UI. Worktrees on `review-1` branch (D:\Capstone, `main` is docs-only).
 - **Scope boundary:** Weeks 1–10 Post-Phase-2 work is the *current system* shown in this deck; **Capstone Project Phase 2** content is added manually by the team and is not shown here.
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2. Complete System Architecture & Pipeline (2 slides)

2.1 End-to-End Pipeline (Post-Phase-2)

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Track A: Scraping (YT IG Reddit) → creators, creator_related_accounts
(relation_type="frequent_collaborator"),
      youtube_channels/videos, instagram_profiles/posts (caption 100-char
truncation), reddit_posts/post_creators
      ↓
Track C: feature_store.py (staging) → scrubbed raw_text + thumbnails +
log_subscriber_count + engagement_rate
```

```

(+ reputation_score=None gap flagged, not fabricated) →
collaboration edges (resolved handles, ambiguous dropped)
+ co_occurs_with (reddit_post_creators junction, 1,414 real) -
backend/app/feature_store.py:179/243
↓
Track B: ml/schema.py CREATOR_CATEGORIES
(athlete/team/league/fitness_influencer/lifestyle_influencer/other)
→ build_real_hetero_data.py → PyG HeteroData → GAIL (gail_model.py,
training.py)
→ checkpoint models/gail_checkpoint.pt c6488a6
↓
Track C: spillover.py:get_spillover_batch (single GAT forward, cached) +
fusion.py:57
final_score = (0.4·spillover + 0.3·sentiment + 0.3·feature)·100 +
risk_adj, CI=hw·100·w1
→ scores.py / recommendations (budget hard,
region/demographic/product soft)
↓
Track D: frontend Next 16 (brand-input → dashboard → explainability →
monitoring) via NEXT_PUBLIC_API_BASE_URL=:8000

```

2.2 Repository & Runtime

- main a4b3bed docs+plans only; code lives on track-a 8429d97 / track-b 69157df / track-c deaf630 / track-d eb8dc98 merged into review-1 e4b8477 (D:\Capstone). No opencode.json .
- **Verified live (2026-08-27):** 259 creators, 54 pair rows, 170 edges, 340 collaborations, 1,414 co-occurrences (pair_count.py canonical). Pooler DB only — direct IPv6 host fails WinError 10051 . REST verify CAPSTONE_NEXT_STEPS.md:259 .

3. Post-Phase-2 Development Overview (1 slide)

- **What changed after Capstone Project Phase 2:** All current modules were built *Post-Phase-2* — ingestion hardening, pair_count canonical, feature_store staging, real co_occurs_with , GAIL checkpoint c6488a6, spillover live (spillover_basis trained/inferred/isolated/placeholder), fusion with honest small-N CI, tiered placeholder cost + empty-query explainability (259 considered / 116 region / 128 product drops for Athletic water bottle demo), and frontend demo polish (clickable handles, humanized hover, simplified explainability).
- **Why it matters:** System moved from no-op stub (influencers.py pre-2026-08-09) to an honest, end-to-end brand-queryable pipeline — the baseline from which Review 2 will improve.

4. Core Module 1 — Data Ingestion & Feature Staging (2 slides)

4.1 Ingestion (Track A)

- Orchestrator `scripts/ingestion/orchestrator.py:1` (1460 lines) + `pair_count.py:1` canonical; 7 migrations in `supabase/migrations/`. Constraints honored (`creators.category CHECK`, `relation_type="frequent_collaborator"`, both endpoints must exist, `instagram_profiles.creator_id` nullable, sheet `1UX9K3...VPQ` never write `approval_status`).
- **Gaps acknowledged:** Instagram 12-post first-paint cap + caption `og:description` workaround (`CAPSTONE_NEXT_STEPS:645`), shortcode base64 timestamp backfill, YouTube||anything safe else IG→Reddit sequentially.

4.2 Feature Store (Track C, `feature_store.py:1`)

- **Does:** `build_creator_features()` — `log_subscriber_count`, `engagement_rate` (pooled likes+comments/reach, Reddit excluded — no denominator), `category_one_hot` (exact order `ml/schema.py`), `raw_text` scrubbed (`text_processing.py`), `thumbnail_urls` — staged for Track B CLIP/BERT (Weeks 9-10, not embedded here). Caps `_MAX_CONTENT_ITEMS=20` .
 - **Does NOT fabricate:** `reputation_score` always `None` until source column defined (open cross-track, `API_CONTRACTS.md`); `sponsors` edges empty until `POST /labeling/run` .
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5. Core Module 2 — Graph Construction & GAIL Spillover (3 slides)

5.1 Graph (Track B `ml/schema.py` , `build_real_hetero_data.py`)

- **Nodes:** `creators` (`CREATOR_CATEGORIES` 6-way one-hot + metadata dim). **Edges:** `collaborates_with` (resolved `creator_related_accounts` → handles, ambiguous lebron duplicate dropped), `co_occurs_with` (shared `reddit_post_creators` — 1,414 real, e.g. PV Sindhu↔Saina 5 posts via `r/badminton`), `sponsors` (pending labeling).

5.2 GAIL Model (`gail_model.py` , `training.py` , `scripts/train_holdout_round3.py`)

- **Training:** Holdout LOO over `N=10` (throwaway models, no checkpoint), `WeightedSAGEConv` + `spillover_head`, propensity saturates `1.000` (`CAPSTONE:795`) → small-N honest.
- **Inference:** `spillover.py:get_spillover_batch` single GAT forward, cached; `spillover_score` nominal 0-1 but live can exceed (e.g. Virat 21.6) → clamped in fusion;

spillover_basis ∈ trained (labeled, hw≈3.28→±13) / inferred (graph-connected unlabeled, 1.6×→±21, min 0.25) / isolated (degree 0 → 0.5 never inferred) / placeholder.

5.3 Checkpoint & Spillover Live

- models/gail_checkpoint.pt c6488a6 (backend/models/ copy). Verified: GET /scores/c4b20dc1... Virat trained 21.61→100, abdevilliers17 inferred 1.19→77, _bungy_lover_ isolated 0.5→50 [40-60].
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6. Core Module 3 — Fusion & Recommendation Engine (2 slides)

6.1 Fusion Layer (fusion.py:57)

- final_score = w1·spillover + w2·sentiment + w3·feature with w1=0.4, w2=0.3, w3=0.3 (only w1 real; w2/w3 placeholder 0.5 until Temporal CAPSTONE_NEXT_STEPS:822). RISK_THRESHOLD=0.3 → -10 pts.
- Honest CI:** hw = t_{0.975,df}·σ·sqrt(1+1/N) with σ=sqrt(mse) mse=1.84 df=N-2 t=2.306@N=10 (gail/inference.py table); margin = hw·100·w1 clamped [0,100] — even trained ±13. Only spillover variance modeled; w2/w3 fixed → final CI = hw·100·w1. PLACEHOLDER_CONFIDENCE_MARGIN=8 fallback.

6.2 Recommendation Engine (routers/influencers.py:160 , API_CONTRACTS.md)

- POST /recommendations:** creators ≤1000 → resolve YouTubeChannel/InstagramProfile → get_spillover_batch → filter → rank by final_score.
 - Filters:** budget hard (estimated_cost=reach·rate_for(category) tiered athlete 0.60 / fitness 0.35 / lifestyle 0.40 / team 0.45 / other 0.50 — was flat 0.5, tracking/TASK2_ANALYSIS.md; unknown reach not filtered); platform_preference hard (handle exists); target_region/demographic/product_category soft — only exclude on *confirmed mismatch* when signal exists (youtube_channels.country/description, instagram_profiles.bio, creator.category), matching any(k in combined for k in keywords) where keywords are ≥3-char words (_extract_keywords:66, _keyword_overlap:72) — not whole-phrase.
 - Demo fix:** Now returns {explanation?,counts?} when results==[] (Athletic water bottle / 5000000 / India → 259 considered, 15 budget, 116 region, 128 product).
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7. Additional Important Modules (2 slides)

7.1 Sponsorship Labeling & Rate Heuristic

- `app/labeling.py` (`POST /labeling/run`) populates `is_sponsored / sponsorship_raw_matches / brands` (`source='sponsorship_mention'` only) — Track C writes; `is_bot_flagged/bot_score` = Track B. Cost is a tiered placeholder (no rate-card yet) shown as (placeholder rate) ; full table migration path `tracking/TASK2_ANALYSIS.md:44 option 2 ( brand_rate_cards )`.

7.2 Monitoring & Alerts (`routers/alerts.py`)

- `POST /alerts` (auth `X-API-Key` when set) + `GET /alerts ( ? creator_id&include_resolved )`. `riskalert smoke-test id2 LeBron high "Weeks 7-8 propagation-field smoke test" has been resolved ( resolved=true )` so `GET /alerts → []` → Monitoring shows No alerts yet (expected until Temporal ships; `propagated_from_creator_id` ready for weeks 14-15 sentiment propagation).

8. Web Application / System Demonstration (2 slides)

8.1 Brand-Input → Dashboard → Explainability → Monitoring (Track D `frontend/src/app/`)

- **Brand Input:** `product_category` (athlete vs Athletic water bottle substring gap demoed), `budget` (₹5M tiered), `target_region / demographic` → `POST /recommendations` → `sessionStorage.recommendationResult` (hydrated via `useSyncExternalStore useStoredRecommendationResult.ts:9`).
- **Dashboard:** Three states — `!result` → “No query yet”, `results==[]` → new empty Explainability (`explanation + counts`), results present → ranked cards with `SpilloverBadge` (emerald trained / violet inferred / zinc isolated) now humanized via `tooltipCopy(basis, influencerName)` (“Estimated for Priya from similar creators...”), **clickable handles** (`instagram→youtube→reddit` → `instagram.com/...` / `youtube.com/@...` / `reddit.com/r|user/...`), no placeholder hints (“placeholder 0.5” , “raw GAIL...” , “confidence 0-100” removed, only `final_score + basis`).
- **Explainability:** Simplified from technical dump → human line `Final score X - Spillover Y pts (40%) + Sentiment Z pts (30%) + Features W pts (30%) + Estimated range X-Y for <name> ; removed hw≈5.25 , is_mock_data true ... , raw GAIL outside nominal... , Inferred - graph-connected...Wide CI placeholders; graph footer now honest: “Graph data is live (340 + 1,414 edges via GET /feature-store/edges/, GAIL c6488a6) but the interactive network-graph is still a UI placeholder”*` (not a data gap).

- **Monitoring:** `getAlerts()` + `getCreators()` name resolution; empty state is the honest current.

8.2 Handoff & Env

- `NEXT_PUBLIC_API_BASE_URL=http://127.0.0.1:8000`, `npm run dev :3000` / `npm run build` / `npm run lint`, Docker at `%LOCALAPPDATA%\Programs\DockerDesktop...\docker.exe`. Env pooler only.
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9. Retrained Model & Updated Results (1 slide)

- **Current model:** GAIL checkpoint `c6488a6` (trained $N=10$, throwaway LOO) is the *retrained* model since Review 1 staging — `POST /scores/compute` and `get_spillover_batch` both live. **Updated results:** Trained exemplars (Virat 21.61 raw $\rightarrow 100$, best fusion), inferred neighbors (AB de Villiers 1.19 $\rightarrow 77$, PV Sindhu 5.05, Ellyse Perry 1.86) and isolated (`_bungy_lover_` 0.5 $\rightarrow 50$) correctly separated by `spillover_basis`. *Not yet retrained* (planned for Review 2, §16): Temporal $\rightarrow$ real `sentiment_risk_score`, CLIP/BERT $\rightarrow$ real `creator_feature_score` + calibration of `w1-3` and backfill of `fusionscore` (removes `is_mock_data:true`).
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10. Metrics & Evaluation (2 slides)

10.1 Current Metrics (honest small-N)

- Fusion CI coverage via `hw·100·w1` (trained ± 13 , inferred ± 21 , placeholder ± 10 , clamped); `RISK_THRESHOLD 0.3` risk adjustment; recommendation filter counts (`considered` / `dropped_by_*`) surfaced as explainability; no sales ROI — engagement-per-rupee proxy only (`PROJECT_PLAN.md §5`).

10.2 Gaps Flagged (not hidden)

- `sentiment_risk_score` & `creator_feature_score` both 0.5 $\rightarrow$ 15.0 pts each per `fusion.py:9` (Temporal 0%); `reputation_score` None; `is_mock_data:true` until `fusionscore` backfilled; `DEFAULT_RECENCY_DAYS=1095` ceiling not widened.
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11. Baseline Comparison (1 slide)

- **Cost:** flat 0.5 vs tiered `CATEGORY_RATE` (current) vs future `brand_rate_cards` table — rank stability within budget bracket is the baseline; `dropped_by_budget` counts validate.

- **Spillover:** heuristic prior (no graph) vs GAT trained/inferred/isolated separation — small-N honest CI is the improvement; propensity 1.000 collapse documented.
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12. Results / Error Analysis / Important Observations (2 slides)

12.1 What Works

- End-to-end brand query is live; keyword-overlap soft filters correctly keep no-signal creators (prevents empty sets during ramp); collaboration edges now real (handle-resolved, ambiguous dropped) and co-occurrence genuine (r/badminton multi-creator).

12.2 What Fails / Lessons

- **“Enabled ≠ reachable”** — pooler vs IPv6, Docker non-standard path, `opencli` daemon starvation; **“Never trust a guessed handle”** — 4/5 guessed mapped to fan accounts; **Silent zeros** from string/table mismatches (`relation_type` literal); **Verify consumer not writer** — filter string vs table; **Instagram 100-char caption truncation** + `og:description` join workaround; **Shortcode base64 timestamp** 99.4% <72 h median 0.5 d backfill.
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13. Team Contributions (1 slide)

- **Track A Data/Infra** — ingestion `orchestrator.py:1` + `pair_count.py:1` + `supabase/migrations` (7) → 259/1711 edges facts.
 - **Track C Fusion+Backend** — `feature_store.py:1`, `fusion.py:57`, `routers/scores.py:1`, `API_CONTRACTS.md:1` → live scoring & recommendations + demo polish batch.
 - **Track B ML-Core** — `ml/schema.py:1`, `gail_model.py:1`, `training.py:1`, `GRAPH_SCHEMA.md:1`, `train_holdout_round3.py:1` → HeteroData + GAIL c6488a6.
 - **Track D Frontend+App** — `frontend/src/app/` (5 routes), `lib/api.ts:1`, `SpilloverBadge.tsx` → UI + humanized demo polish.
 - Each track also maintains `tracking/track-*.md` + `TASK2_ANALYSIS.md` for replayability.
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14. Remaining Work — What Is Unfinished Right Now? (1 slide)

What is unfinished right now? (complements §15–18 Review 2 Plan, no duplication)

- **Fusion realism:** `sentiment_risk_score` & `creator_feature_score` still 0.5 (Temporal 0% CAPSTONE_NEXT_STEPS:822, CLIP/BERT weeks 9-10 pending) → Dashboard shows 15.0 pts each; `fusionscore` not backfilled → `is_mock_data:true`.
- **Graph UI:** Data is live (340/1,414 edges + GAIL), but **interactive network-graph / D3 force-graph & Granger-causal posting-time insight** not rendered in `frontend/src/app/explainability/` (footer placeholder).
- **Cost model:** Tiered `CATEGORY_RATE` is still a code-level heuristic; no persisted `brand_rate_cards` table, no engagement-adjusted rate.
- **Calibration:** `w1-3` still 0.4/0.3/0.3 placeholder, not calibrated on held-out outcomes; CI derived only from spillover variance.
- **Evaluation maturity:** No NDCG/similar rank eval, no ablation of `w2/w3`, no robustness/Error analysis beyond filter counts; Monitoring `riskalert` propagation (`propagated_from_creator_id`) not exercised.
- **Integration/perf:** No comprehensive `pytest` (69 backend 49) + load test + Docker next build gate in CI.

15. Review 2 — Current State to Next Phase (Review 2 Slide 1)

During Review 2, the project moves from an honest placeholder-fusion system to a calibrated, evaluated final system.

Visual: Current System → Remaining Development → Review 2 → Final System (swimlane, see ORCHESTRATION.md timeline weeks 11-13+)

- **Already completed (Post-Phase-2 work):** Ingestion hardening (54 pair rows / 1,711 edges truth), `co_occurs_with` real, PyG HeteroData, GAIL checkpoint `c6488a6` with `trained/inferred/isolated` honest CI, `feature_store` staging, live APIs (`POST /recommendations` with `{explanation,counts}`, `GET /scores/*`, `GET /feature-store/edges/*`, `GET /alerts`), web app `brand-input→dashboard→explainability→monitoring` with demo polish (clickable `instagram→youtube` handles, humanized `Inferred for <name>...` tooltip, simplified explainability, empty-query guidance).
- **Currently functional:** A brand can query (`athlete + ₹5M + India`) and receive a ranked, 0–100 scored list with honest basis badges and per-filter drop explainability — even on a small-N model.
- **Remains incomplete (see §14):** Real sentiment & creator-feature scores, calibrated fusion, persisted rate-card, graph visualization & causal insight, and comprehensive evaluation.

- **Review 2 phase will focus on:** Completing the two remaining model branches (Temporal & CLIP/BERT), calibrating the fusion layer, wiring the graph visualization, and **proving** the system via metrics/baselines/ablations — not adding administrative scope.
- **How it builds on the existing system:** Every Review 2 task consumes the current pipeline as input (same 259 creators, same HeteroData, same `get_spillover_batch`) and replaces placeholder `0.5` paths with learned values, so the final system is an evolution, not a rebuild.

16. Review 2 — Planned Development (Review 2 Slide 2)

Prioritized meaningful technical work planned for the Review 2 phase — each row is `What` → `Which module` → `Why` → `Outcome` → `Impact on final system`.

#	What will be developed / improved	Module(s) affected
1	Complete Temporal sentiment-propagation branch — sentiment classifier + propagation over <code>collaborates_with</code> / <code>co_occurs_with</code> → real <code>sentiment_risk_score</code> in <code>[0,1]</code> + <code>risk_alert.propagated_from_creator_id</code>	<code>ml/</code> (new Temporal), <code>backend/</code> <code>backend/app/fusion.py:34</code> , <code>backend/app/routers/alerts.py</code>
2	Complete CLIP + BERT creator-feature pipeline — run <code>raw_text</code> (scrubbed <code>text_processing.py</code>) through BERT and <code>thumbnail_urls</code> through CLIP → real <code>creator_feature_score</code> (+ fill <code>reputation_score</code> once source column is defined)	<code>ml/schema.py:CREATOR_METADATA</code> extraction (Weeks 9-10), <code>backend/app/feature_store.py</code>
3	Calibrate & backfill the Fusion layer — rerun <code>scripts/build_real_hetero_data.py</code> → <code>scripts/train_holdout_round3.py</code> (no longer throwaway) → recalibrate <code>w1/w2/w3</code> (beyond <code>0.4/0.3/0.3</code>) against held-out outcomes, then backfill <code>fusionscore</code> rows for all 259	<code>backend/app/fusion.py:57</code> , <code>backend/app/spillover.py</code>
4	Persist the rate-card & engagement-adjust — migrate tiered <code>CATEGORY_RATE</code> → table <code>supabase/migrations/0008_brand_rate_cards</code> (<code>category</code> , <code>platform</code> , <code>region</code> , <code>cost_per_follower</code>) seeded from marketplace CPM research + multiply by <code>1+0.2*(engagement_rate-median)</code> from <code>feature_store.compute_engagement_rate</code>	<code>backend/app/routers/influencer.py</code> <code>supabase/migrations/</code>

#	What will be developed / improved	Module(s) affected
5	Wire the graph & causal UI — D3 force-graph in <code>frontend/src/app/explainability/</code> consuming already-live <code>`GET /feature-store/edges/collaborations</code>	co-occurrence (340/1,414) + granger-causal posting-time/lag (GAIL stable)
6	Web-app polish & perf — tighten brand-input validation, keep SpilloverBadge humanized copy (Estimated for <name>...), keep empty-query explanation + counts ; cache <code>get_spillover_batch</code> result, handle <code>instagram_profiles.creator_id=null</code> comment rows (CAPSTONE_NEXT_STEPS:399) without clobber	<code>frontend/src/app/brand-input</code> <code>frontend/src/lib/useStoredRe</code> <code>backend/app/spillover.py</code>

Planned work for the Review 2 phase is strictly the above — no “Capstone Project Phase 2” content and no new scrape beyond the tiered rate seed.

17. Review 2 — Planned Evaluation & Validation (Review 2 Slide 3)

*Review 2 is not simply about adding features, but about **proving** the completed system performs effectively — each evaluation ties directly to a current limitation (§14).*

Evaluation	Metrics / Setup	Baseline / Comparison	Con limit
Rank quality	NDCG@5/10 + Spearman vs engagement-per-rupee proxy (not sales PROJECT_PLAN \$5 ROI note) on time-split holdout (pre-2025 train → 2025 test within DEFAULT_RECENCY_DAYS=1095)	Flat-rate vs tiered vs persisted rate-card; GAT-spillover vs heuristic spillover; placeholder 0.5 vs learned sentiment/feature (ablation <code>w2=0 / w3=0</code>)	Plac → c flat c cons
Score accuracy & honesty	MSE on held-out <code>spillover_score</code> , CI coverage (does 95% CI actually cover ~95% after $hw=t \cdot \sigma \cdot \sqrt{(1+1/N)}$ recalibration?),	trained ± 13 vs inferred ± 21 vs isolated ± 10 bands; pre/post-calibration <code>w1-3</code>	Sma toda tight hona

Evaluation	Metrics / Setup	Baseline / Comparison	Con limit
	risk_adjustment precision when sentiment<0.3		
Ablation	Drop each feature group (log_subscriber_count , engagement_rate , category one-hot , raw_text , thumbnails) + each edge type (collaborates_with vs co_occurs_with)	Δ NDCG + Δ MSE; edge-less GAIL vs graph-aware GAIL	rep Non flagg its a
Qualitative / error analysis	Spot-check r/badminton PV Sindhu*Saina 5 posts co-occurrence; per-creator explainability audit (does Estimated range X-Y for <name> match basis?); filter-drop audit (dropped_by_product top reason for Athletic water bottle)	Manual vs keyword-overlap _keyword_overlap (≥ 3 chars) — not whole-phrase; before/after brand_rate_cards on same ₹5M India query	Soft crud stern — fi rate
Alerts / robustness	Alert precision/recall once Temporal propagation exercises propagated_from_creator_id ; red-team short/empty product_category (“x”) must not drop everyone (_keyword_overlap empty→ False not mismatch)	Smoke-test id2 LeBron (now resolved) vs real propagation alerts; duplicate-handle lebron disambiguation test	Mon smo prov prop
System & integration	pytest tests/ -q (69) + pytest backend/tests/ -q (49) green; npm run build (Next 16) passes; GET /alerts <100 ms, POST /recommendations (1000 creators) p95 <500 ms, get_spillover_batch single GAT forward cached	sqlite:///./fusion_backend.db fallback vs pooler; cmd wrapper vs WMI detached host survival (see tracking/track-c)	Perf cre mus scal

The Review 2 phase will report these numbers in §10–12 of the deck, not just ship code.

18. Review 2 — Expected Final System (Review 2 Slide 4)

After the Review 2 work is completed, the system will be:

- **Fully integrated:** Ingestion (A) → Feature Store (C staging + B embeddings) → Graph (HeteroData: collaborators 340 + co-occurrence 1,414 + sponsors) → GAIL (spillover) + Temporal (sentiment) → Fusion (calibrated w1-3) → API (recommendations / scores / feature-store / alerts with real propagation) → Web App (brand-input → ranked dashboard with clickable instagram→youtube handles + D3 graph → explainability with per-creator Estimated for → monitoring with real alerts) .
- **Model-improved:** No longer one real branch — both remaining branches learned (sentiment_risk_score & creator_feature_score replace 0.5), fusion weights calibrated on held-out data, fusionscore backfilled (no is_mock_data), CI narrowed where data allows but still honest where N is small.
- **Web-app at demo parity:** npm run build + Docker (docker compose up) one-command demo; empty-query explanation + counts stays, tiered cost becomes rate-card v1 with engagement adjustment, graph visualization is interactive not placeholder, Granger-causal posting-time insight is scaffolded.
- **Evaluation-mature:** NDCG/similar rank metric + baselines + ablations + qualitative error analysis + system perf report — ready for final demonstration and hand-off to the tracks via tracking/track-*.md master prompts.

*Visual: pipeline evolution diagram — **Current (muted, placeholder 0.5 / flat 0.5 / table UI)** on left → arrow “Review 2 (complete branches + calibrate + visualize + evaluate)” → **Planned (highlighted, learned scores / calibrated / graph UI / validated)** on right. Do not claim planned improvements as already achieved.*

19. Review 2 — Priorities / Roadmap (Review 2 Slide 5)

The roadmap reflects real dependencies from ORCHESTRATION.md / CAPSTONE_NEXT_STEPS.md:963 and tracking/ replay order — it is a separate slide per request.

Phase order for the Review 2 phase (planned work for Review 2):

1. **Complete remaining core functionality** — finish Temporal (sentiment) + CLIP/BERT (features) so raw_text / thumbnail_urls stop being staged (feature_store.py:158/160) and become embeddings.
2. **Integrate modules** — wire embeddings + sentiment into HeteroData (build_real_hetero_data.py) and fusion.py (feed both branches into get_spillover_batch → compute_fusion_score).
3. **Retrain / optimize where required** — rerun train_holdout_round3.py + compute_training_pair_deltas.py (only if confident it helps demo — Rule 4), recalibrate w1-3 and hw margins; mark a commit point (user Rule 1) before shared re-use.

4. **Conduct comprehensive evaluation** — NDCG / Spearman / MSE / CI coverage + baseline comparisons (flat vs tiered vs rate-card; graph-aware vs graph-less) + ablation $w_2=0/w_3=0$ / edge-type drop.
5. **Perform error / robustness analysis** — `_keyword_overlap` false-drop audit, `estimated_cost` bracket stability check (₹5M India query as canary), duplicate-handle & `creator_id=null` safety drill, short-query red-team.
6. **Improve the web application** — wire D3 force-graph in `explainability`, keep Dashboard humanized polish (`SpilloverBadge` `tooltipCopy`), add rate-card traceability; `npm run build` gate.
7. **Final testing & validation** — full `pytest` green, `npm run lint/build`, pooler REST re-verify (`CAPSTONE_NEXT_STEPS:259`), Docker smoke, `pair_count.py` canary (259/340/1414 not regressed).
8. **Prepare final system / demo** — freeze `review-1` → `review-2` branch, update `tracking/track-*.md` master prompts per change (so each track can replay), and rehearse the end-to-end brand query demo.

During Review 2 the team will stay on the `review-1` branch until the roadmap's final freeze, restart backends/frontends via subagents for verification (`uvicorn :8000` / `next dev :3000`), and only commit on your say-so.
